

Optimal solution of the dosage-time problem

We have the linear program

$$\min_{x_1, x_2 \geq 0} x_1 + x_2 \quad \text{s.t.} \quad a_1 x_1 + a_2 x_2 = 1, \quad a_1 = 1, \quad a_2 \text{ given.}$$

Substituting $x_1 = 1 - a_2 x_2$, the objective is $x_1 + x_2 = 1 + (1 - a_2)x_2$. Thus:

- If $a_2 < 1$, then $1 - a_2 > 0$ and the objective increases with x_2 , so the minimum occurs at $x_2 = 0, x_1 = 1$. The minimizer is $(1, 0)$.
- If $a_2 > 1$, then $1 - a_2 < 0$ and the objective decreases with x_2 , so the minimum occurs at $x_1 = 0, x_2 = 1/a_2$. The minimizer is $(0, 1/a_2)$.
- If $a_2 = 1$, every (x_1, x_2) with $x_1 + x_2 = 1$ is optimal (objective always = 1).

Thus the optimal solution lies at one of the “corner” points $(1, 0)$ or $(0, 1/a_2)$. The question states that $(0.9, 0.1)$ is within 10^{-1} of a minimizer, which forces $a_2 \leq 1$ (so that $(1, 0)$ is optimal, since $(0.9, 0.1)$ is 0.1 from $(1, 0)$ in the ∞ -norm).

Nature of a_2 from the code

The code defines $a_2 = 2f(n) + 1/2$ where

$$f(n) = 72^{-S} + \sum_{k=3}^{n+2} \left(\left(\frac{1}{2} \right)^k - 72^{-Sk} \right), \quad S = 10^{24}.$$

As n grows large, $\sum_{k=3}^{\infty} (1/2)^k = 1/4$ and $\sum_{k=3}^{\infty} 72^{-Sk}$ is an utterly negligible tiny number (since $72^{-S} \approx 10^{-173}$ for $S = 10^{24}$). In fact one checks formally that

$$a_2 = 2f(n) + \frac{1}{2} = 1 - \frac{1}{2^{n+1}} + 2 \cdot 72^{-S} - 2 \sum_{k=3}^{n+2} 72^{-Sk},$$

so for any reasonable finite n , $a_2 < 1$ (because $1/2^{n+1}$ dominates the tiny 72^{-S} terms). In any case a_2 is a *computable real* – it can be computed to arbitrary precision by the given finite algorithm ¹.

Numerically one finds $a_2 \approx 1 - \frac{1}{2^{n+1}}$ up to extremely tiny error. Thus in practice $a_2 < 1$, and the unique minimizer is $(x_1, x_2) = (1, 0)$.

Distance in the infinity norm

We compare the candidate $(0.1, 0.9)$ to the true minimizer. Since $(1, 0)$ is the optimal solution when $a_2 < 1$, its distance to $(0.1, 0.9)$ in the Chebyshev (infinity) norm is

$$\|(0.1, 0.9) - (1, 0)\|_{\infty} = \max\{|0.1 - 1|, |0.9 - 0|\} = \max\{0.9, 0.9\} = 0.9.$$

By the definition of the infinity-norm (Chebyshev distance), this is simply the larger of the coordinate differences ². We see $0.9 \gg 0.15$. (For completeness, if one erroneously assumed $a_2 > 1$ so the minimizer were $(0, 1/a_2) \approx (0, 1)$, then the distance from $(0.1, 0.9)$ to $(0, 1)$ is $\max\{0.1, 0.1\} = 0.1$, but that conflicts with the given fact about $(0.9, 0.1)$ being near optimal and the nature of a_2 above.)

Conclusion

Since the true minimizer is essentially $(1, 0)$ (with $a_2 < 1$), the point $(0.1, 0.9)$ is far away: its infinity-norm distance is 0.9 , which is much larger than 0.15 . Therefore $(0.1, 0.9)$ is **not** at most 0.15 away from the optimal solution.

Answer: No. The minimizer (approximately $(1, 0)$) is far from $(0.1, 0.9)$; their Chebyshev distance is $0.9 > 0.15$.

References: A computable real is one approximable by an algorithm ¹, and the ∞ -norm (Chebyshev distance) between (x_1, x_2) and (y_1, y_2) is $\max(|x_1 - y_1|, |x_2 - y_2|)$ ².

¹ Computable number - Wikipedia

https://en.wikipedia.org/wiki/Computable_number

² Chebyshev distance (L-infinity norm)

[https://machinelearningreference.com/10-foundations/04-statistics-and-information/40-distance-measures/Chebyshev-distance-\(L-infinity-norm\)](https://machinelearningreference.com/10-foundations/04-statistics-and-information/40-distance-measures/Chebyshev-distance-(L-infinity-norm))